

Conformal Uncertainty Quantification for Mammographic Malignancy Classification

Knowing When the Model Doesn't Know

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Dataset: CBIS-DDSM — 3,568 biopsy-confirmed mammograms

Method: Split conformal prediction (LAC / APS / RAPS + Mondrian)

Coverage target: $\alpha = 0.05 \rightarrow 95\%$ guarantee

A DenseNet-121 mammography classifier is wrapped with split conformal prediction — turning uncalibrated scores into **label sets guaranteed to contain the true diagnosis 95 % of the time**, automatically flagging ambiguous cases for radiologist review.

1 Introduction

Breast cancer is one of the most prevalent cancers worldwide. Mammographic screening is the primary early-detection tool, but radiological interpretation is time-consuming, error-prone, and subject to inter-observer variability.

Deep learning classifiers have shown strong performance on medical-image tasks, yet they have a critical weakness in clinical deployment: they output a **single label with an overconfident probability**, providing no principled signal of when the prediction should be distrusted.

This project addresses that gap. We train a standard DenseNet-121 classifier on biopsy-confirmed mammograms (CBIS-DDSM) and wrap it with **split conformal prediction** — a post-hoc uncertainty quantification method that converts raw classifier scores into **prediction sets with a distribution-free coverage guarantee**, with no assumptions on the data distribution or model architecture.

2 The Problem

Our classifier's raw confidence is off by **ECE = 0.205** — a 20-point gap between stated confidence and actual accuracy. This overconfidence is a known property of softmax networks, not a flaw in this specific model.

What screening actually needs: predictions with a *stated, guaranteed error rate* — not a number that merely looks like a probability. A confidently wrong “benign” is a missed cancer.

THE COVERAGE GUARANTEE

$$\Pr(Y_{\text{true}} \in C(X)) \geq 1 - \alpha \quad (\alpha = 0.05 \rightarrow 95\% \text{ coverage})$$

When the set holds **one label** the model is confident and the case can be auto-reported. When it holds **both labels** the model is honestly flagging ambiguity — that case routes to a radiologist.

3 Approach & Pipeline

Backbone: DenseNet-121 pretrained on ImageNet, fine-tuned on CBIS-DDSM for binary (benign / malignant) classification. All splits are at the **patient level** to prevent data leakage and preserve the conformal guarantee.

Train 1,951

Val 352

Calibration 595

Test 670

The **calibration split** is the element unique to conformal prediction: never seen during training, used solely to compute the threshold $\hat{q}$ that delivers the coverage guarantee.

THRESHOLD & SET CONSTRUCTION

$$\hat{q} = [(n+1)(1-\alpha)]/n \text{ quantile of } \{1-p_{\text{true},i}\} \text{ on cal set}$$
$$C(x) = \{k : p_k(x) \geq 1 - \hat{q}\}$$

4 The Three Non-Conformity Scores

Three scoring rules are compared. They share the same calibration framework but differ in how the nonconformity score is defined, leading to different set sizes.

Method	Score definition	Set size $\pm$ sd	Singleton %
LAC	$1 - p_{\text{true}}$	1.694 $\pm$ 0.027	30.6 %
APS	Cumulative sorted mass	1.720 $\pm$ 0.028	28.0 %
RAPS	APS + set-size penalty	1.714 $\pm$ 0.029	28.6 %

All three hold $\geq 95\%$ coverage. **LAC gives the tightest sets on a binary task** — consistent with theory that RAPS and APS earn their added complexity only when the label space is large. Comparing all three, rather than asserting one, is itself a methodological contribution.

5 Why Our Numbers Are Trustworthy

Before touching any mammogram, all three methods were validated on **synthetic data with known statistics** across 200 independent trials at $K = 2$ and $K = 5$. This separates a *correct implementation* from a *fortunate result*.

LAC K=2 0.951 ✓

APS K=2 0.955 ✓

RAPS K=2 0.953 ✓

References: Angelopoulos & Bates (2021) · Romano, Sesia & Candès (2020) · Angelopoulos et al. (2021, RAPS) · Sadinle et al. (2019, LAC) · Sawyer-Lee et al. (2016, CBIS-DDSM)

6 Results

Base classifier (12 epochs, single-view): val accuracy 0.733 · test accuracy 0.643 · **AUC 0.708**. Single-view CBIS-DDSM is genuinely hard; the uncertainty layer is the contribution.

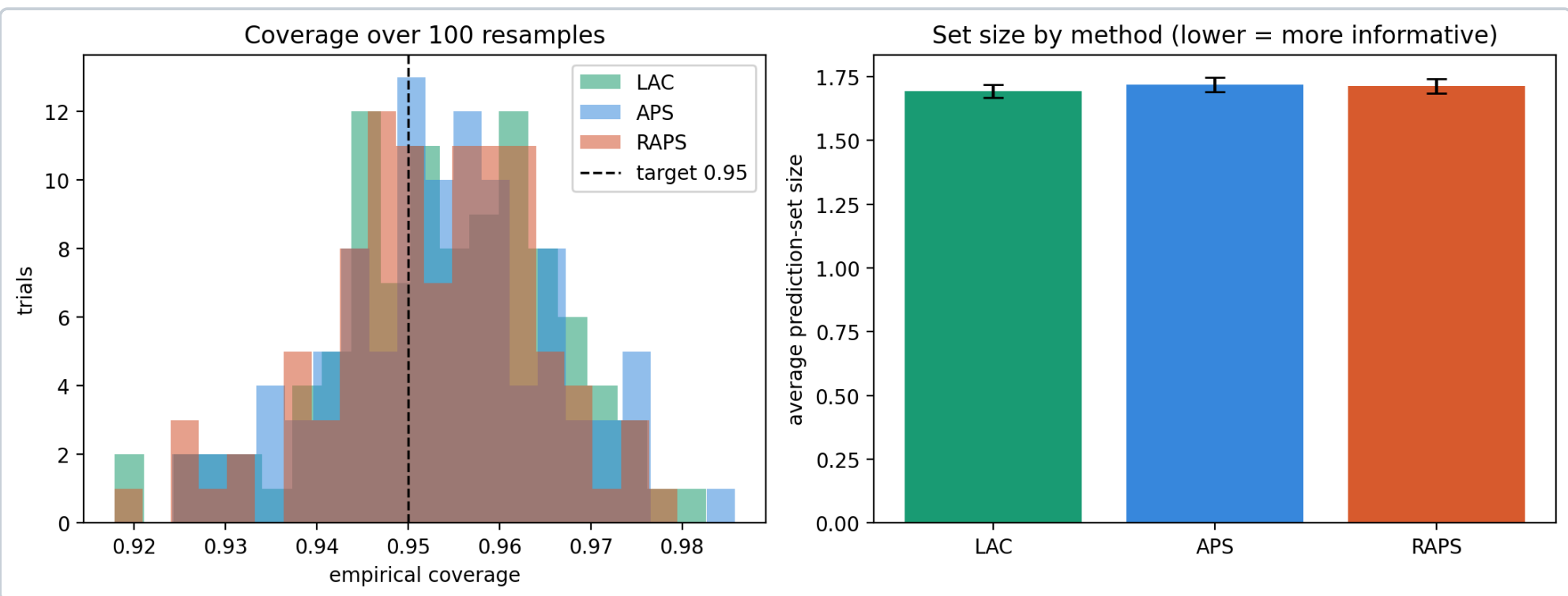


Figure 1. Coverage distributions and average set sizes across 100 resamples.

Left: all three methods cluster tightly around the 0.95 target (dashed) across 100 independent calibration/test resamples — the guarantee holds, not just on one lucky split. **Right:** LAC (teal) achieves the smallest average set size (1.694), resolving the most cases to a single confident label. This is the expected result for a binary task and validates the method selection.

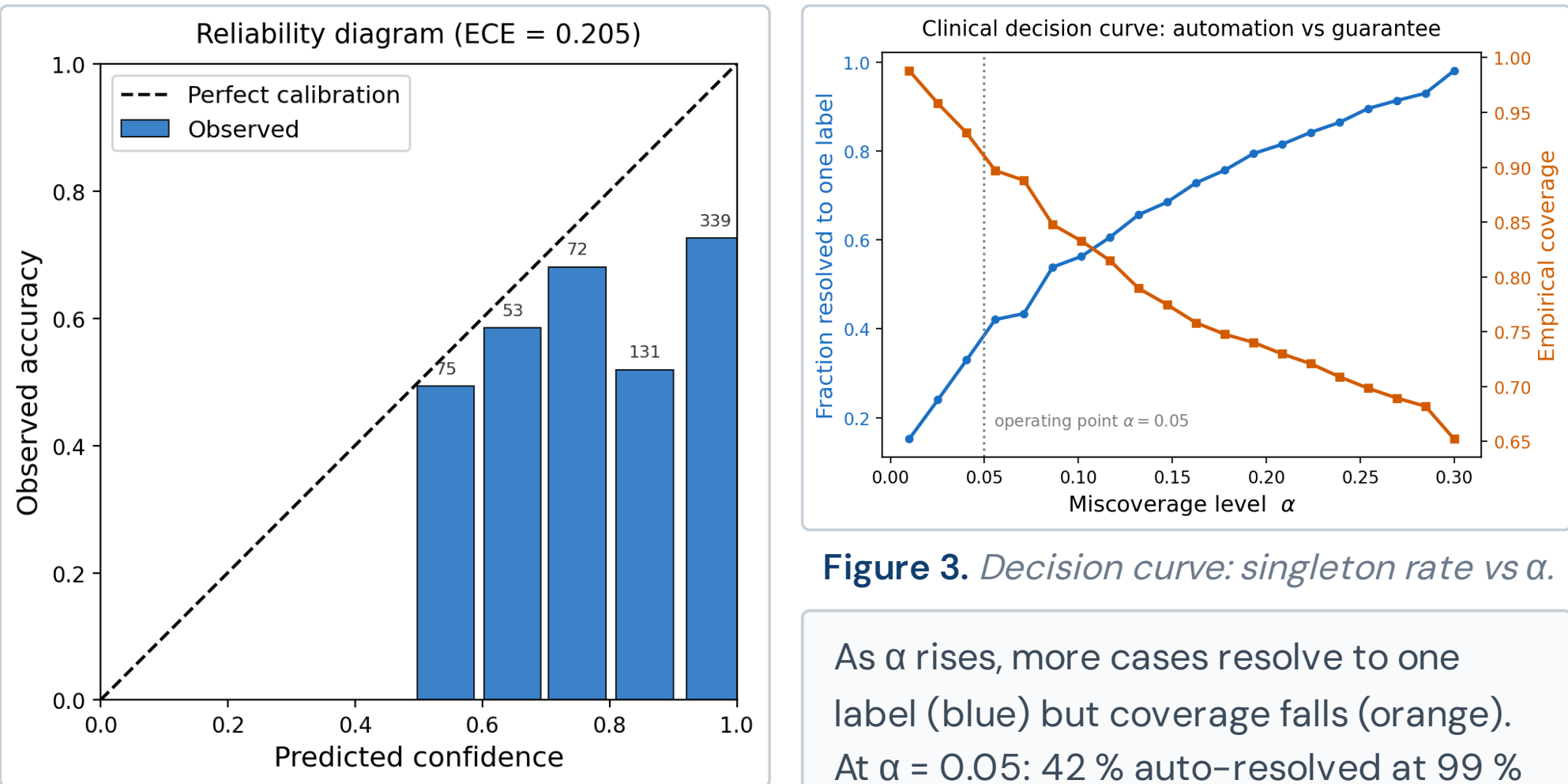


Figure 2. Reliability diagram (ECE = 0.205).

Bars below the diagonal = over-confident model. ECE = 0.205 quantifies the 20-point gap between stated confidence and actual accuracy. This is the concrete motivation for conformal prediction.

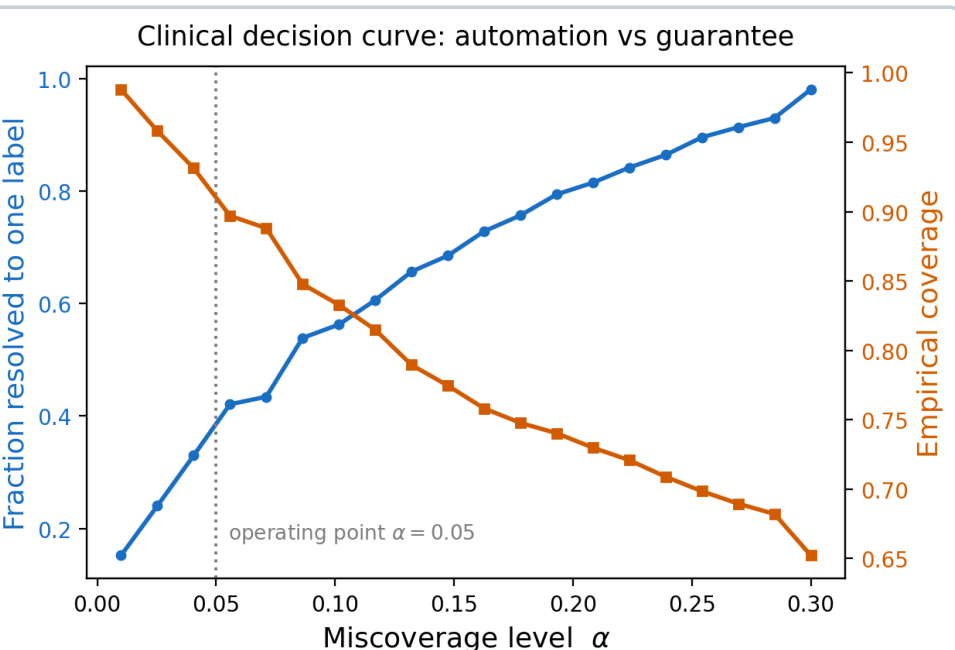


Figure 3. Decision curve: singleton rate vs α .

As α rises, more cases resolve to one label (blue) but coverage falls (orange). At $\alpha = 0.05$: 42 % auto-resolved at 99 % coverage. This is the operating-point curve a clinical team would use.

7 Real Mammograms with Prediction Sets

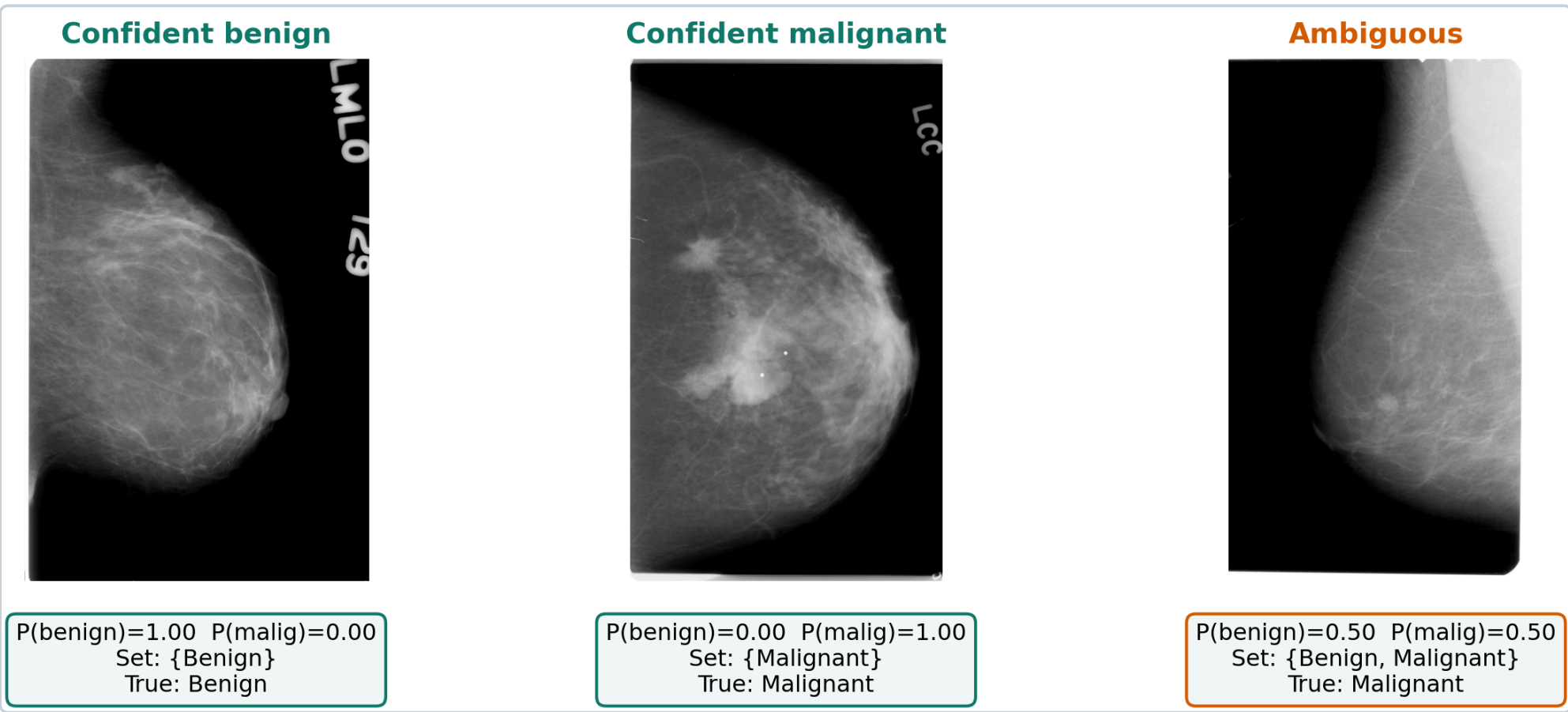


Figure 4. Three test-set mammograms with LAC prediction sets ($\alpha = 0.05$).

Three real cases from the held-out test set show the system working in practice. **Left (Confident benign):** $P(\text{malignant}) = 0.00$ — the set collapses to {Benign} and the case can be auto-reported without radiologist input. **Centre (Confident malignant):** $P(\text{malignant}) = 1.00$ — the set is {Malignant} with full confidence; the case is flagged for immediate follow-up. **Right (Ambiguous):** $P(\text{benign}) = P(\text{malignant}) = 0.50$ — both labels are included; the system honestly flags its uncertainty. The true label is Malignant: this is exactly the case that must not be auto-reported, and the prediction set correctly prevents that.

The **set size is the uncertainty signal**: a singleton means confidence; a two-label set means honest admission of ambiguity and automatic escalation to expert review.

8 Class-Conditional Coverage: A Fix

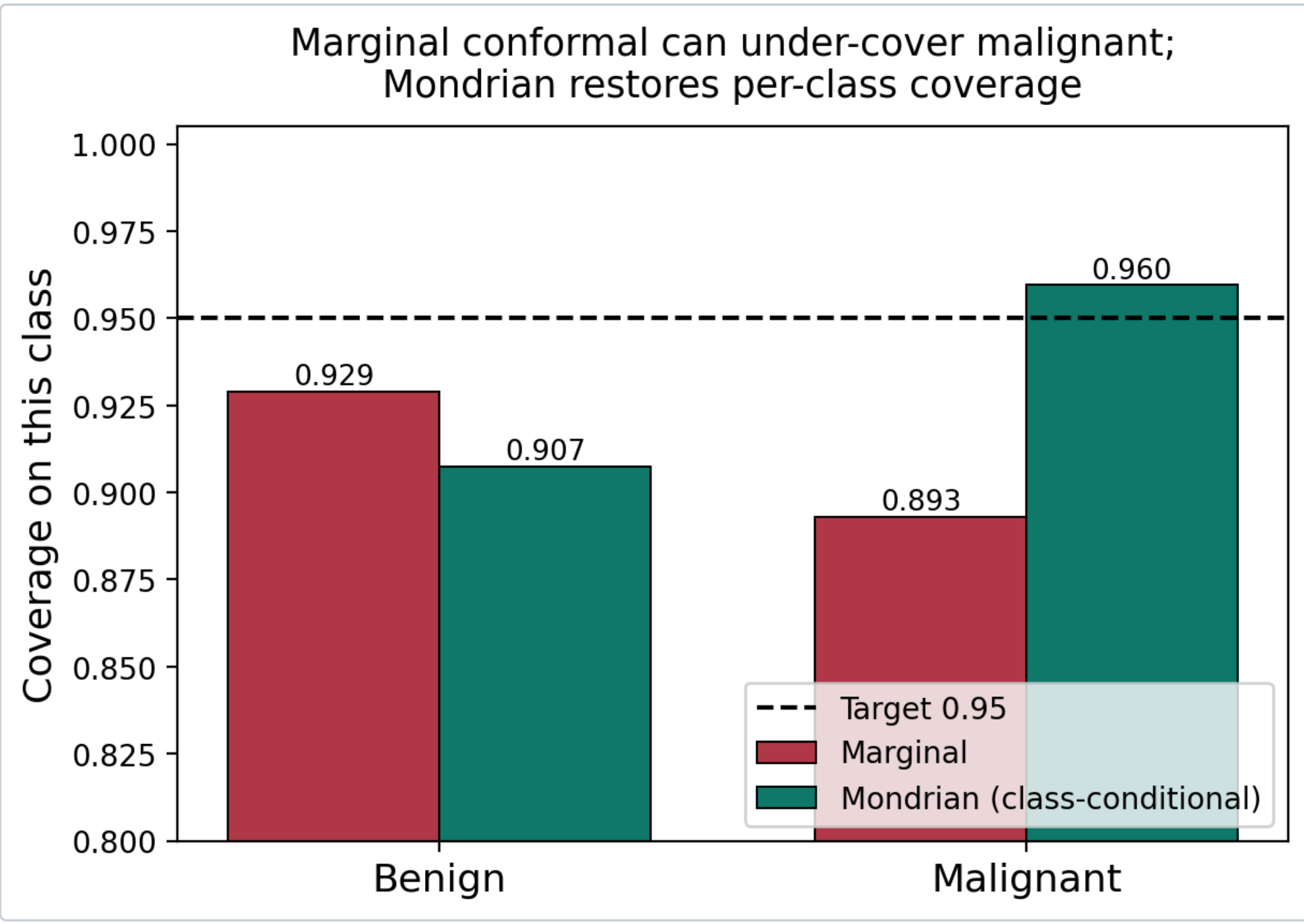


Figure 5. Marginal vs Mondrian per-class coverage.

Red bars (Marginal): standard split conformal holds 95 % overall but under-covers malignant at 0.893, meaning 11 % of true cancers fall outside the prediction set — the most dangerous failure mode in screening. **Teal bars (Mondrian):** calibrating a separate threshold per class restores malignant coverage to 0.960 by construction. Mondrian conformal achieves per-class guarantees at the cost of using only the per-class calibration count; with 595 calibration points the estimate is stable.

Key clinical finding: marginal conformal is insufficient in imbalanced screening settings. Mondrian conformal is the correct tool when per-class safety guarantees are required — and it is the tool that should be deployed.

9 Conclusions

- The guarantee holds empirically.** All three methods achieve $\geq 95\%$ coverage across 100 resamples, verified on synthetic data before any real data was used — a correct implementation, not a fortunate result.
- LAC is most efficient on a binary task** (set size 1.694 vs 1.720/1.714), reproducing the theoretical prediction that RAPS earns its complexity only with large label spaces.
- Marginal conformal under-covers malignant (0.893).** Mondrian per-class calibration restores it to 0.960. This is a clinical distinction, not an academic one: 11 % of missed cancers is unacceptable in screening.
- ECE 0.205 provides concrete evidence** that raw softmax confidence cannot be trusted, motivating the uncertainty layer with a measurable fact.
- The decision curve** turns α from a technical knob into a quantified operating-point choice for clinical teams.

10 Future Work

- Multi-view bilateral fusion.** The Siamese DenseNet-121 architecture with gated cross-view attention is designed and implemented; the conformal layer transfers unchanged to its outputs.
- BI-RADS multi-class grading.** The conformal code already handles $K > 2$ (verified at $K = 5$); extending to five-class BI-RADS is a one-line label change.
- Cross-dataset generalisation.** Deployment on VinDr-Mammo requires weighted or Mondrian conformal variants to handle distribution shift while preserving the per-class coverage guarantee.
- Temperature scaling.** Reducing the base classifier's ECE (currently 0.205) before applying conformal prediction would tighten prediction sets further while keeping the guarantee intact.

11 Summary & Code Repository

“We take a breast-cancer image classifier and wrap it in a mathematical procedure that turns its overconfident guesses into honest, guaranteed statements of uncertainty — so the system knows when to defer to a radiologist.”

The full pipeline — data loading, DenseNet-121 training, conformal calibration (LAC / APS / RAPS), Mondrian fix, and all four extended analyses — is available as a reproducible Kaggle notebook. The standalone conformal.py module and synthetic-data verification script are included so the guarantee can be checked independently of any model.



Scan for the full repository

Replace this placeholder with your QR code image

Dataset: CBIS-DDSM (awsaf49/Kaggle) · Backbone: DenseNet-121 (torchvision) · Conformal: split LAC / APS / RAPS + Mondrian